# MSHA Expanded Ontology Development

## Block 1 - 1,000-Narrative Stress Test

Generated October 14, 2026 at 20:58

Model: Qwen/Qwen2.5-7B-Instruct
GPU: Tesla T4
Development narratives: 1,000
Cause categories: 18
Factor categories: 17
Prevention categories: 17
Successful classifications: 991
Records requiring review: 608

Purpose:
Stress-test the draft ontology, prompt, compatibility rules, evidence checks, and review logic before freezing the system for the unseen 2,000-record analysis.

Important:
This is a development dataset. Its results may be used to improve the ontology and prompt. Block 2 must exclude every Block 1 record ID.

# Executive Summary

This report expands the initial 100-narrative Qwen trial to a 1,000-narrative development sample.

The development sample was selected across five injury-severity groups and broad time periods. Only records routed as acute accidents were included in the primary ontology stress test.

The model assigned one primary cause, one primary contributing factor, and one preliminary prevention theme to each narrative. It also returned supporting evidence, confidence values, support levels, and review notes.

The report evaluates whether the draft ontology is sufficiently broad, whether any categories are overused or unused, how often the model returns insufficient information, whether evidence excerpts can be located in the narrative, and whether contributing factors align with their proposed prevention themes.

The results are intended to support final ontology and prompt decisions. They are not a final test of model accuracy because these development records may influence the final design.

# Development Methodology

1. The existing accidents DataFrame was used directly.

2. Blank, extremely short, and exact duplicate narratives were removed.

3. Records were routed into acute accidents, occupational illnesses, noninjury accidents, and natural or nonoccupational events.

4. The primary development sample included acute-accident narratives.

5. Sampling was balanced across five severity groups and distributed across broad time periods where records were available.

6. Qwen2.5-7B-Instruct was loaded locally in four-bit mode.

7. The model was restricted to the draft cause, factor, and prevention ontology.

8. Responses were normalized and validated with Pydantic.

9. Supporting evidence was checked using exact, normalized, and token-overlap methods.

10. Factor-prevention combinations were checked against deterministic compatibility rules.

11. Records with low support, missing evidence, unusual relationships, truncation, or fallback categories were flagged for review.

12. Development record IDs were saved separately so Block 2 can exclude every record used in this ontology-development stage.

# Severity Sampling Summary

| Severity Group | Available | Target | Selected |
|---|---|---|---|
| Fatality | 1090 | 200 | 200 |
| Permanent disability | 2447 | 200 | 200 |
| Days away | 176835 | 200 | 200 |
| Restricted or transferred | 42614 | 200 | 200 |
| No lost time or other | 4043 | 200 | 200 |

# Development Sample by Severity and Time Period

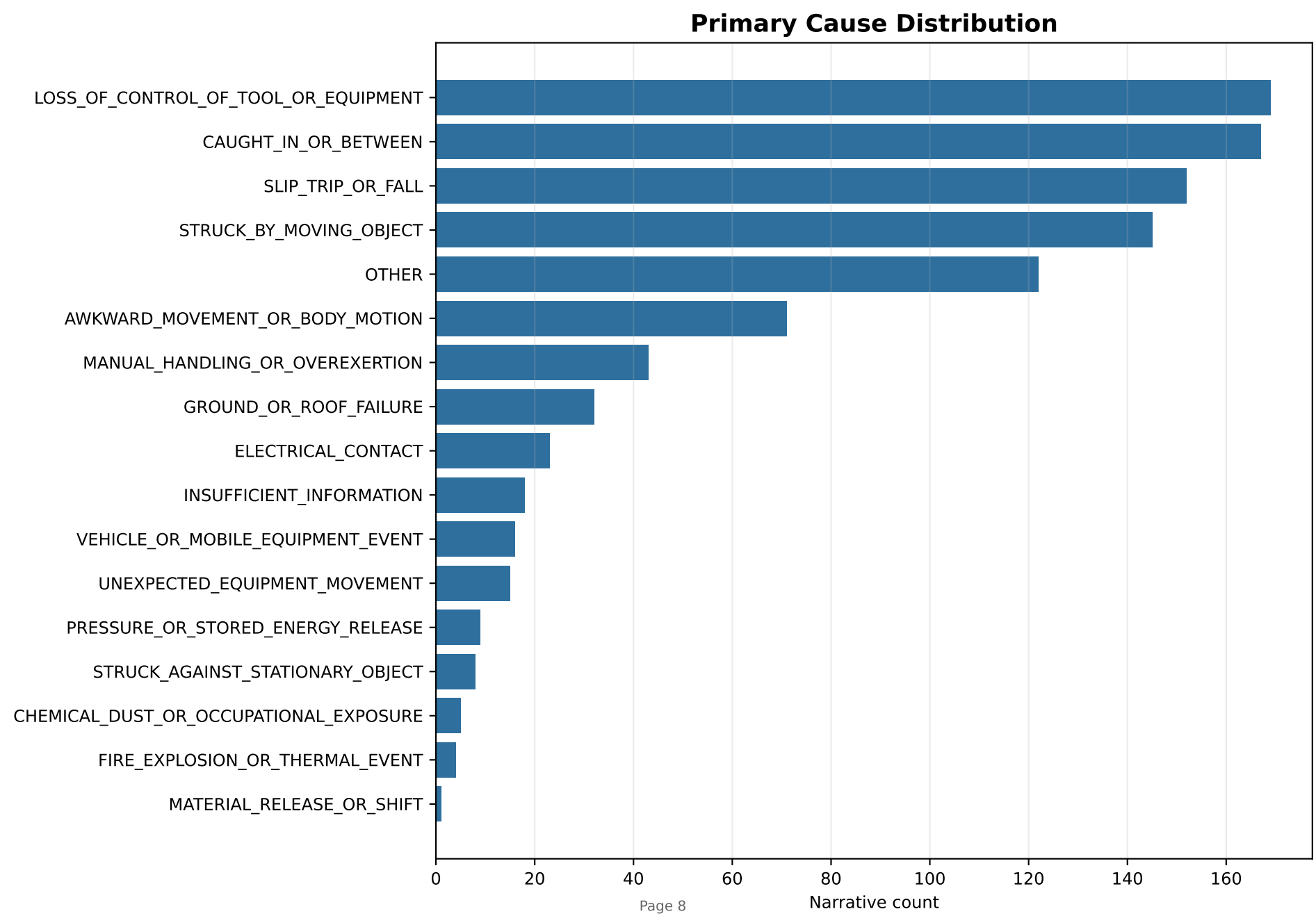
| Severity Group | Early: 2000-2008 | Middle: 2009-2017 | Recent: 2018-2026 |
|---|---|---|---|
| Days away | 67 | 67 | 66 |
| Fatality | 66 | 67 | 67 |
| No lost time or other | 67 | 67 | 66 |
| Permanent disability | 67 | 66 | 67 |
| Restricted or transferred | 67 | 67 | 66 |

# Processing and Quality Summary

| Measure | Value |
|---|---|
| Development narratives | 1000.0000 |
| Successful classifications | 991.0000 |
| Failed classifications | 9.0000 |
| Schema-valid results | 991.0000 |
| Records requiring review | 608.0000 |
| Cause evidence exact | 761.0000 |
| Cause evidence normalized | 822.0000 |
| Factor evidence exact | 706.0000 |
| Factor evidence normalized | 751.0000 |
| Factor-prevention alignment flags | 540.0000 |
| NOT_STATED factors | 109.0000 |
| Insufficient-information causes | 18.0000 |
| Truncated or near-limit narratives | 11.0000 |
| Mean processing seconds | 17.7200 |

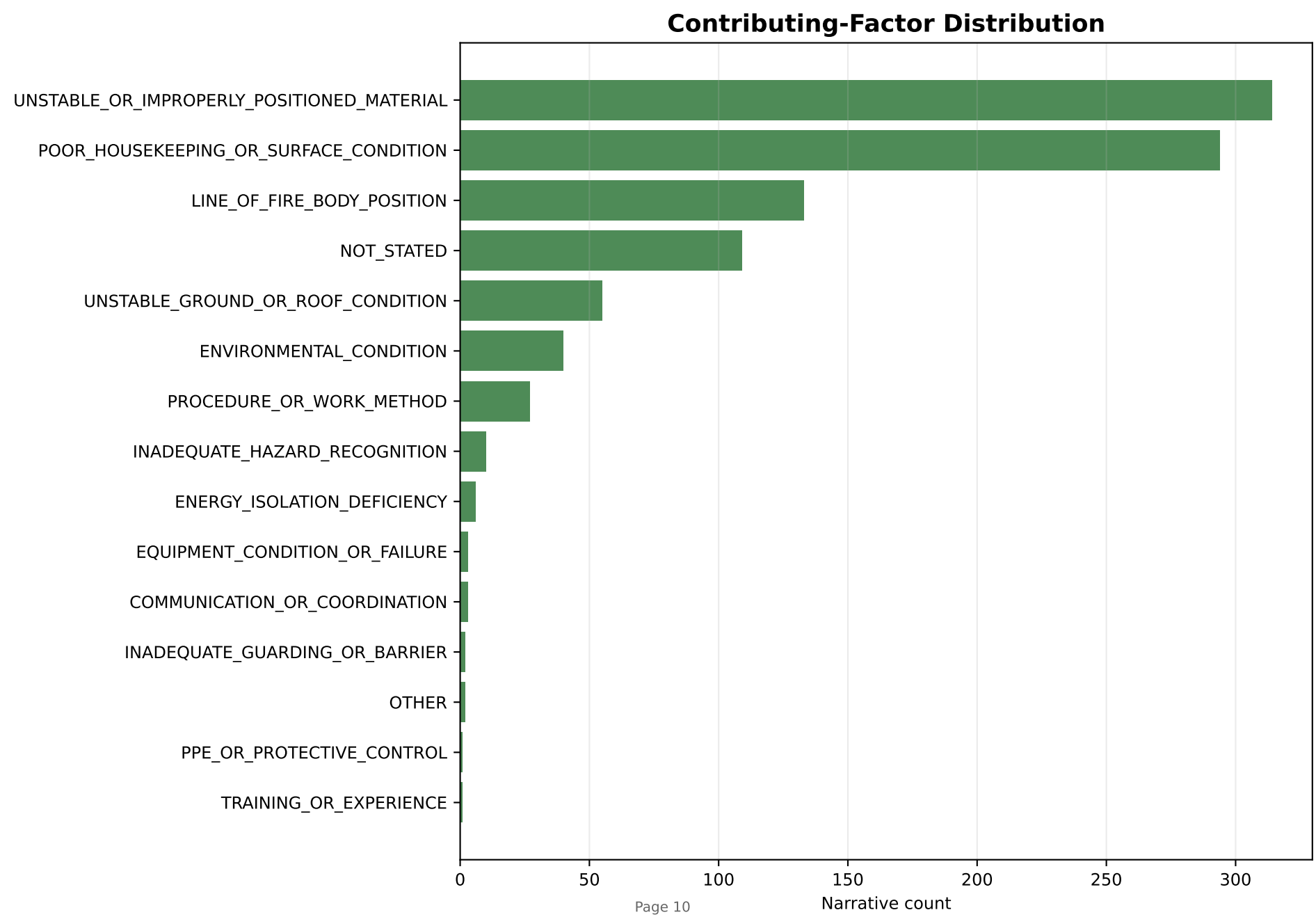
# Primary Cause Distribution

| Primary Cause | Count | Percent |
|---|---|---|
| LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT | 169 | 16.9000 |
| CAUGHT_IN_OR_BETWEEN | 167 | 16.7000 |
| SLIP_TRIP_OR_FALL | 152 | 15.2000 |
| STRUCK_BY_MOVING_OBJECT | 145 | 14.5000 |
| OTHER | 122 | 12.2000 |
| AWKWARD_MOVEMENT_OR_BODY_MOTION | 71 | 7.1000 |
| MANUAL_HANDLING_OR_OVEREXERTION | 43 | 4.3000 |
| GROUND_OR_ROOF_FAILURE | 32 | 3.2000 |
| ELECTRICAL_CONTACT | 23 | 2.3000 |
| INSUFFICIENT_INFORMATION | 18 | 1.8000 |
| VEHICLE_OR_MOBILE_EQUIPMENT_EVENT | 16 | 1.6000 |
| UNEXPECTED_EQUIPMENT_MOVEMENT | 15 | 1.5000 |
| PRESSURE_OR_STORED_ENERGY_RELEASE | 9 | 0.9000 |
| STRUCK_AGAINST_STATIONARY_OBJECT | 8 | 0.8000 |
| CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE | 5 | 0.5000 |
| FIRE_EXPLOSION_OR_THERMAL_EVENT | 4 | 0.4000 |
| MATERIAL_RELEASE_OR_SHIFT | 1 | 0.1000 |


Primary Cause Distribution
LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
CAUGHT_IN_OR_BETWEEN
SLIP_TRIP_OR_FALL
STRUCK_BY_MOVING_OBJECT
OTHER
AWKWARD_MOVEMENT_OR_BODY_MOTION
MANUAL_HANDLING_OR_OVEREXERTION
GROUND_OR_ROOF_FAILURE
ELECTRICAL_CONTACT
INSUFFICIENT_INFORMATION
VEHICLE_OR_MOBILE_EQUIPMENT_EVENT
UNEXPECTED_EQUIPMENT_MOVEMENT
PRESSURE_OR_STORED_ENERGY_RELEASE
STRUCK_AGAINST_STATIONARY_OBJECT
CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE
FIRE_EXPLOSION_OR_THERMAL_EVENT
MATERIAL_RELEASE_OR_SHIFT
0
20
40
60
80
100
120
140
160
Narrative count

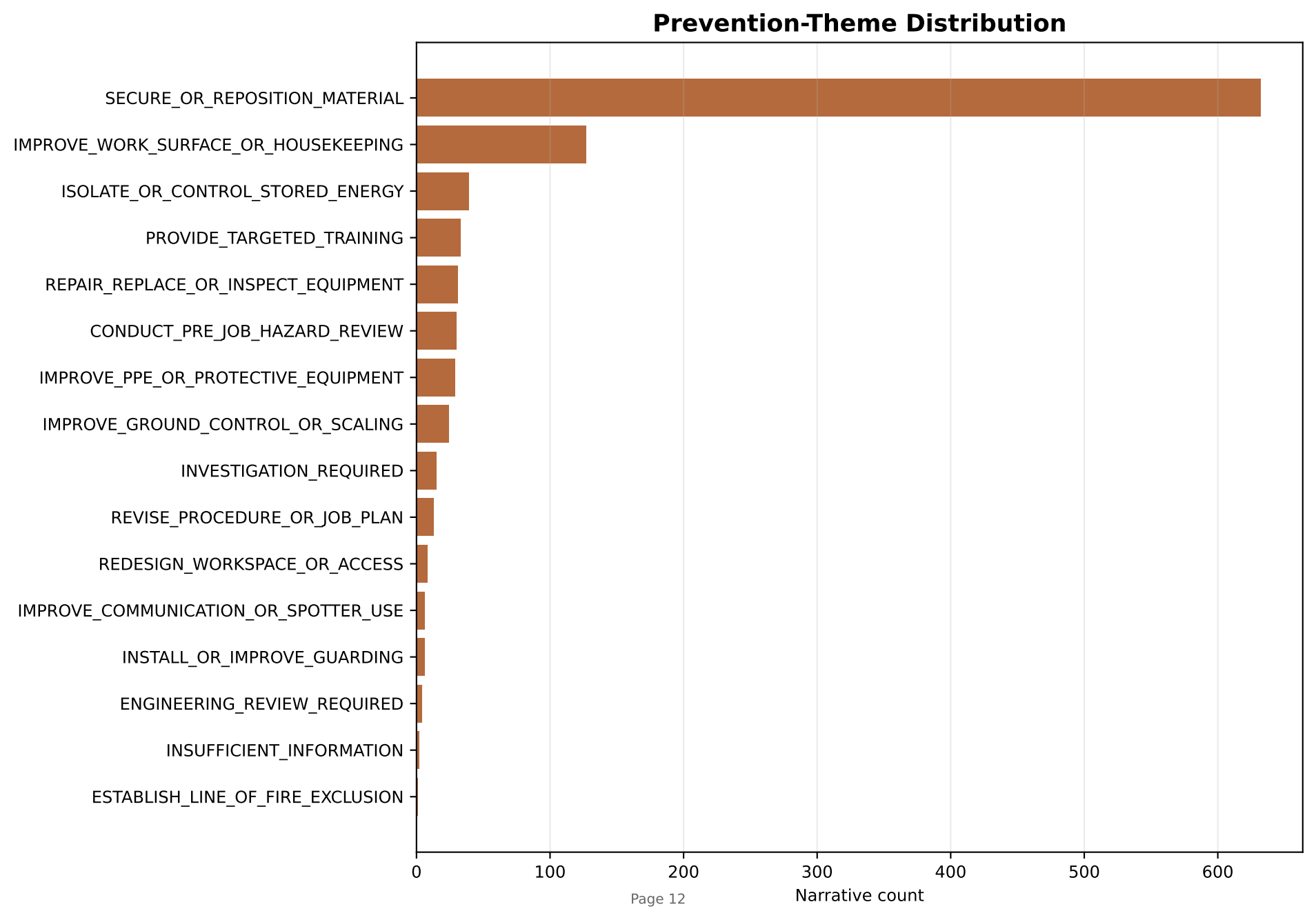
# Contributing-Factor Distribution

| Primary Contributing Factor | Count | Percent |
|---|---|---|
| UNSTABLE_OR_IMPROPERLY_POSITIONED_MATEF | 314 | 31.4000 |
| POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | 294 | 29.4000 |
| LINE_OF_FIRE_BODY_POSITION | 133 | 13.3000 |
| NOT_STATED | 109 | 10.9000 |
| UNSTABLE_GROUND_OR_ROOF_CONDITION | 55 | 5.5000 |
| ENVIRONMENTAL_CONDITION | 40 | 4.0000 |
| PROCEDURE_OR_WORK_METHOD | 27 | 2.7000 |
| INADEQUATE_HAZARD_RECOGNITION | 10 | 1.0000 |
| ENERGY_ISOLATION_DEFICIENCY | 6 | 0.6000 |
| EQUIPMENT_CONDITION_OR_FAILURE | 3 | 0.3000 |
| COMMUNICATION_OR_COORDINATION | 3 | 0.3000 |
| INADEQUATE_GUARDING_OR_BARRIER | 2 | 0.2000 |
| OTHER | 2 | 0.2000 |
| TRAINING_OR_EXPERIENCE | 1 | 0.1000 |
| PPE_OR_PROTECTIVE_CONTROL | 1 | 0.1000 |


Contributing-Factor Distribution
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
LINE_OF_FIRE_BODY_POSITION
NOT_STATED
UNSTABLE_GROUND_OR_ROOF_CONDITION
ENVIRONMENTAL_CONDITION
PROCEDURE_OR_WORK_METHOD
INADEQUATE_HAZARD_RECOGNITION
ENERGY_ISOLATION_DEFICIENCY
EQUIPMENT_CONDITION_OR_FAILURE
COMMUNICATION_OR_COORDINATION
INADEQUATE_GUARDING_OR_BARRIER
OTHER
PPE_OR_PROTECTIVE_CONTROL
TRAINING_OR_EXPERIENCE
0
50
100
150
200
250
300
Narrative count

# Prevention-Theme Distribution

| Prevention Theme | Count | Percent |
|---|---|---|
| SECURE_OR_REPOSITION_MATERIAL | 632 | 63.2000 |
| IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | 127 | 12.7000 |
| ISOLATE_OR_CONTROL_STORED_ENERGY | 39 | 3.9000 |
| PROVIDE_TARGETED_TRAINING | 33 | 3.3000 |
| REPAIR_REPLACE_OR_INSPECT_EQUIPMENT | 31 | 3.1000 |
| CONDUCT_PRE_JOB_HAZARD_REVIEW | 30 | 3.0000 |
| IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | 29 | 2.9000 |
| IMPROVE_GROUND_CONTROL_OR_SCALING | 24 | 2.4000 |
| INVESTIGATION_REQUIRED | 15 | 1.5000 |
| REVISE_PROCEDURE_OR_JOB_PLAN | 13 | 1.3000 |
| REDESIGN_WORKSPACE_OR_ACCESS | 8 | 0.8000 |
| INSTALL_OR_IMPROVE_GUARDING | 6 | 0.6000 |
| IMPROVE_COMMUNICATION_OR_SPOTTER_USE | 6 | 0.6000 |
| ENGINEERING_REVIEW_REQUIRED | 4 | 0.4000 |
| INSUFFICIENT_INFORMATION | 2 | 0.2000 |
| ESTABLISH_LINE_OF_FIRE_EXCLUSION | 1 | 0.1000 |


Prevention-Theme Distribution
SECURE_OR_REPOSITION_MATERIAL
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
ISOLATE_OR_CONTROL_STORED_ENERGY
PROVIDE_TARGETED_TRAINING
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
CONDUCT_PRE_JOB_HAZARD_REVIEW
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
IMPROVE_GROUND_CONTROL_OR_SCALING
INVESTIGATION_REQUIRED
REVISE_PROCEDURE_OR_JOB_PLAN
REDESIGN_WORKSPACE_OR_ACCESS
IMPROVE_COMMUNICATION_OR_SPOTTER_USE
INSTALL_OR_IMPROVE_GUARDING
ENGINEERING_REVIEW_REQUIRED
INSUFFICIENT_INFORMATION
ESTABLISH_LINE_OF_FIRE_EXCLUSION
0
100
200
300
400
500
600
Narrative count

# Cause Ontology Usage

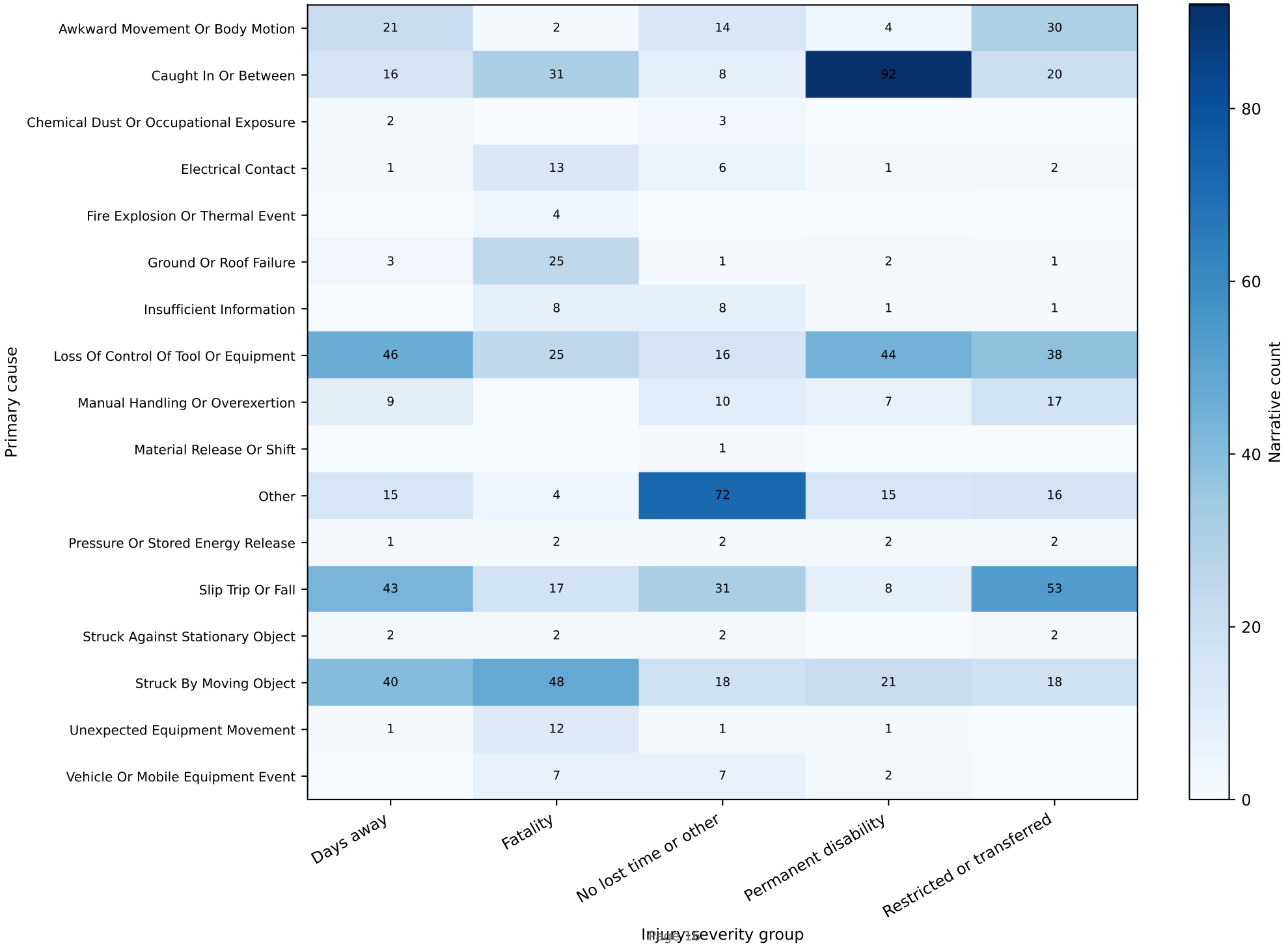
| Category | Count | Percent | Usage Assessment |
|---|---|---|---|
| SLIP_TRIP_OR_FALL | 152 | 15.2000 | Used |
| AWKWARD_MOVEMENT_OR_BODY_MC | 71 | 7.1000 | Used |
| STRUCK_BY_MOVING_OBJECT | 145 | 14.5000 | Used |
| STRUCK_AGAINST_STATIONARY_OBJE | 8 | 0.8000 | Used |
| CAUGHT_IN_OR_BETWEEN | 167 | 16.7000 | Used |
| MATERIAL_RELEASE_OR_SHIFT | 1 | 0.1000 | Very rare |
| GROUND_OR_ROOF_FAILURE | 32 | 3.2000 | Used |
| UNEXPECTED_EQUIPMENT_MOVEMEN | 15 | 1.5000 | Used |
| LOSS_OF_CONTROL_OF_TOOL_OR_EQ | 169 | 16.9000 | Used |
| MANUAL_HANDLING_OR_OVEREXERT | 43 | 4.3000 | Used |
| PRESSURE_OR_STORED_ENERGY_REL | 9 | 0.9000 | Used |
| ELECTRICAL_CONTACT | 23 | 2.3000 | Used |
| FIRE_EXPLOSION_OR_THERMAL_EVEN | 4 | 0.4000 | Very rare |
| VEHICLE_OR_MOBILE_EQUIPMENT_EV | 16 | 1.6000 | Used |
| CHEMICAL_DUST_OR_OCCUPATIONAL | 5 | 0.5000 | Used |
| ERGONOMIC_OR_REPETITIVE_EXPOSU | 0 | 0.0000 | Unused |
| OTHER | 122 | 12.2000 | Used |
| INSUFFICIENT_INFORMATION | 18 | 1.8000 | Used |

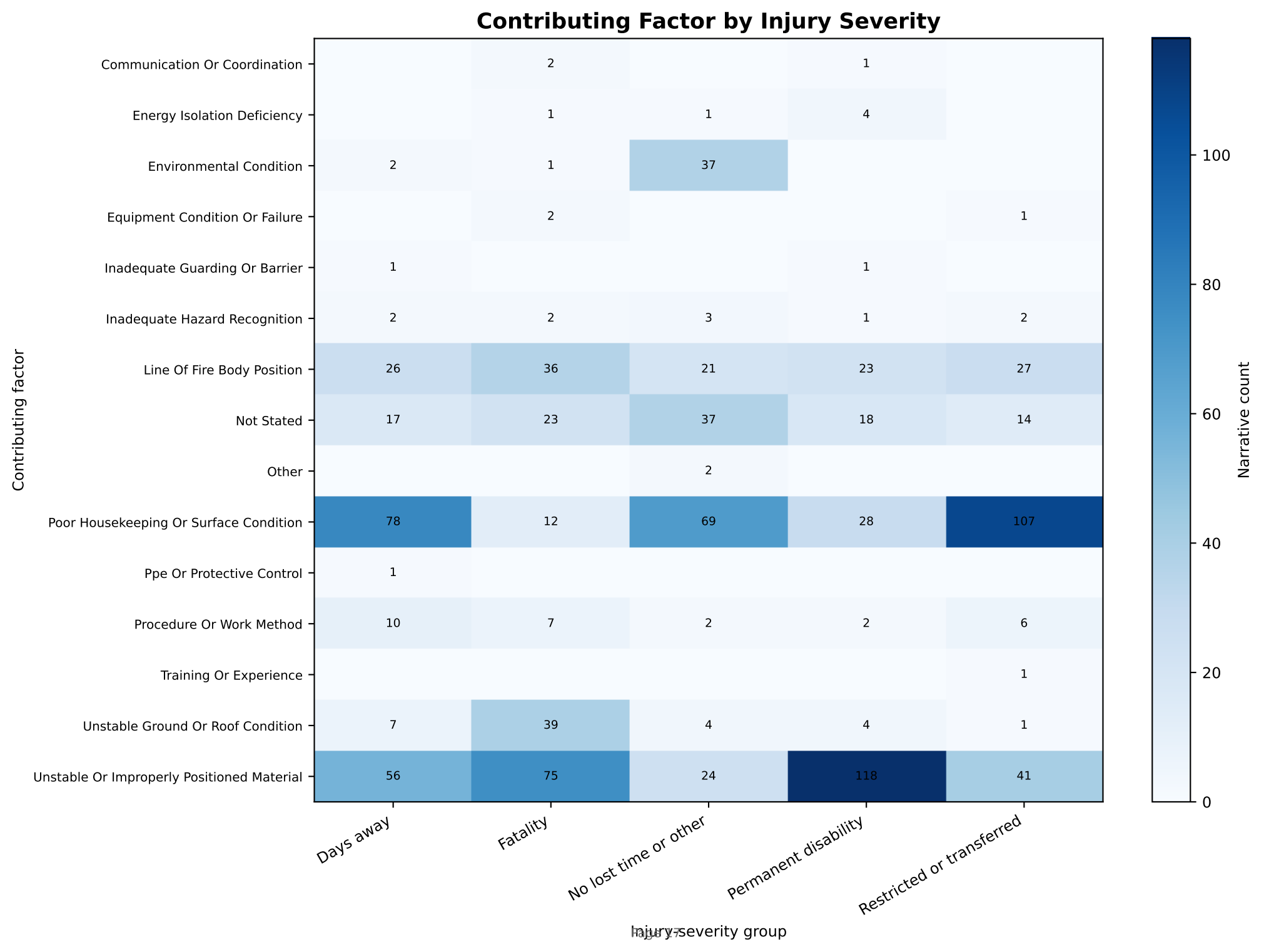
# Factor Ontology Usage

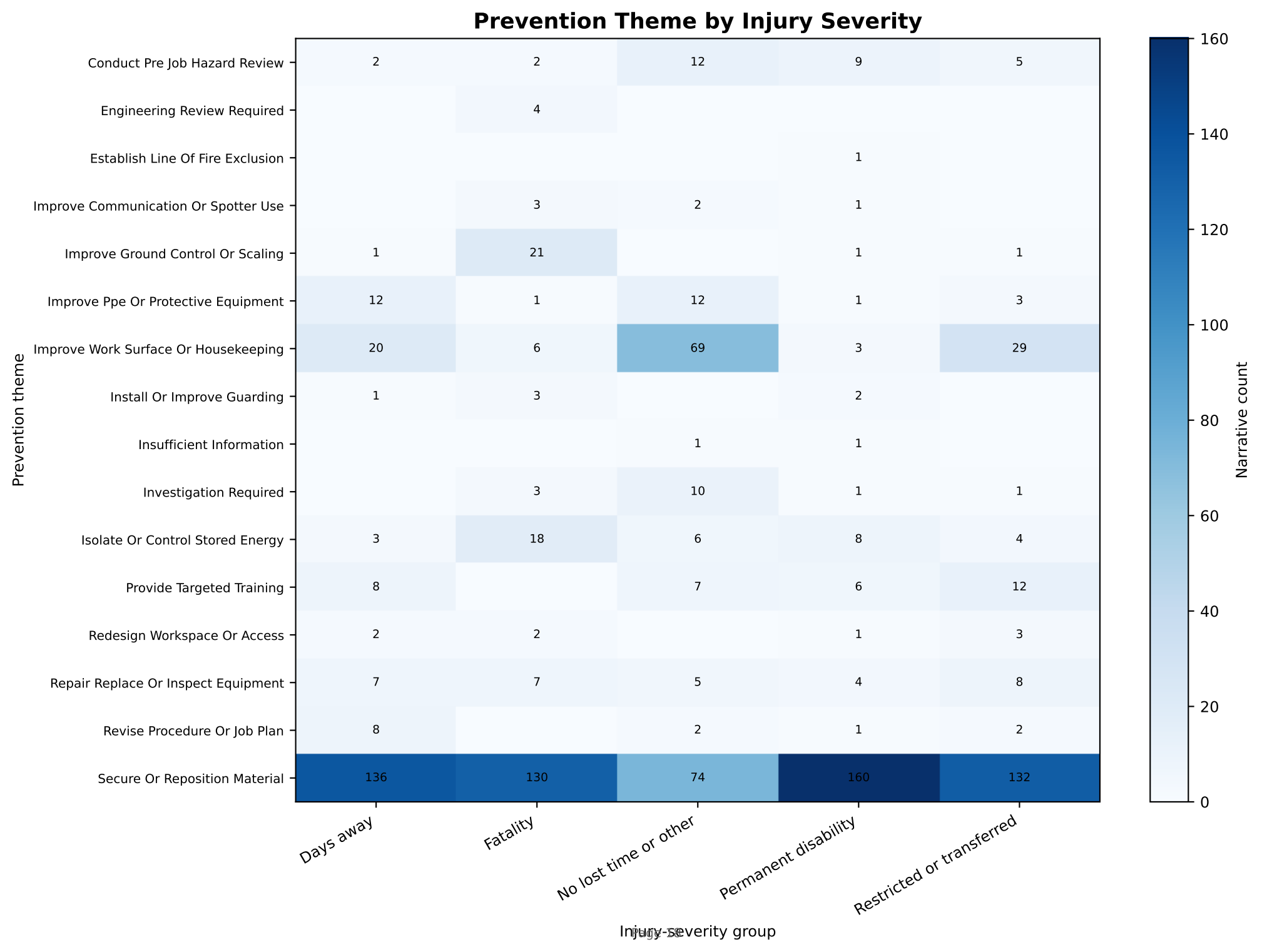
| Category | Count | Percent | Usage Assessment |
|---|---|---|---|
| UNSTABLE_OR_IMPROPERLY_POSITIOI | 314 | 31.4000 | Potentially overused |
| INADEQUATE_SECUREMENT_OR_REST | 0 | 0.0000 | Unused |
| LINE_OF_FIRE_BODY_POSITION | 133 | 13.3000 | Used |
| POOR_HOUSEKEEPING_OR_SURFACE_ | 294 | 29.4000 | Used |
| UNSTABLE_GROUND_OR_ROOF_COND | 55 | 5.5000 | Used |
| INADEQUATE_GUARDING_OR_BARRIE | 2 | 0.2000 | Very rare |
| ENERGY_ISOLATION_DEFICIENCY | 6 | 0.6000 | Used |
| EQUIPMENT_CONDITION_OR_FAILURE | 3 | 0.3000 | Very rare |
| PROCEDURE_OR_WORK_METHOD | 27 | 2.7000 | Used |
| INADEQUATE_HAZARD_RECOGNITION | 10 | 1.0000 | Used |
| COMMUNICATION_OR_COORDINATION | 3 | 0.3000 | Very rare |
| TRAINING_OR_EXPERIENCE | 1 | 0.1000 | Very rare |
| PPE_OR_PROTECTIVE_CONTROL | 1 | 0.1000 | Very rare |
| ENVIRONMENTAL_CONDITION | 40 | 4.0000 | Used |
| WORKSPACE_OR_ACCESS_LIMITATION | 0 | 0.0000 | Unused |
| OTHER | 2 | 0.2000 | Very rare |
| NOT_STATED | 109 | 10.9000 | Used |

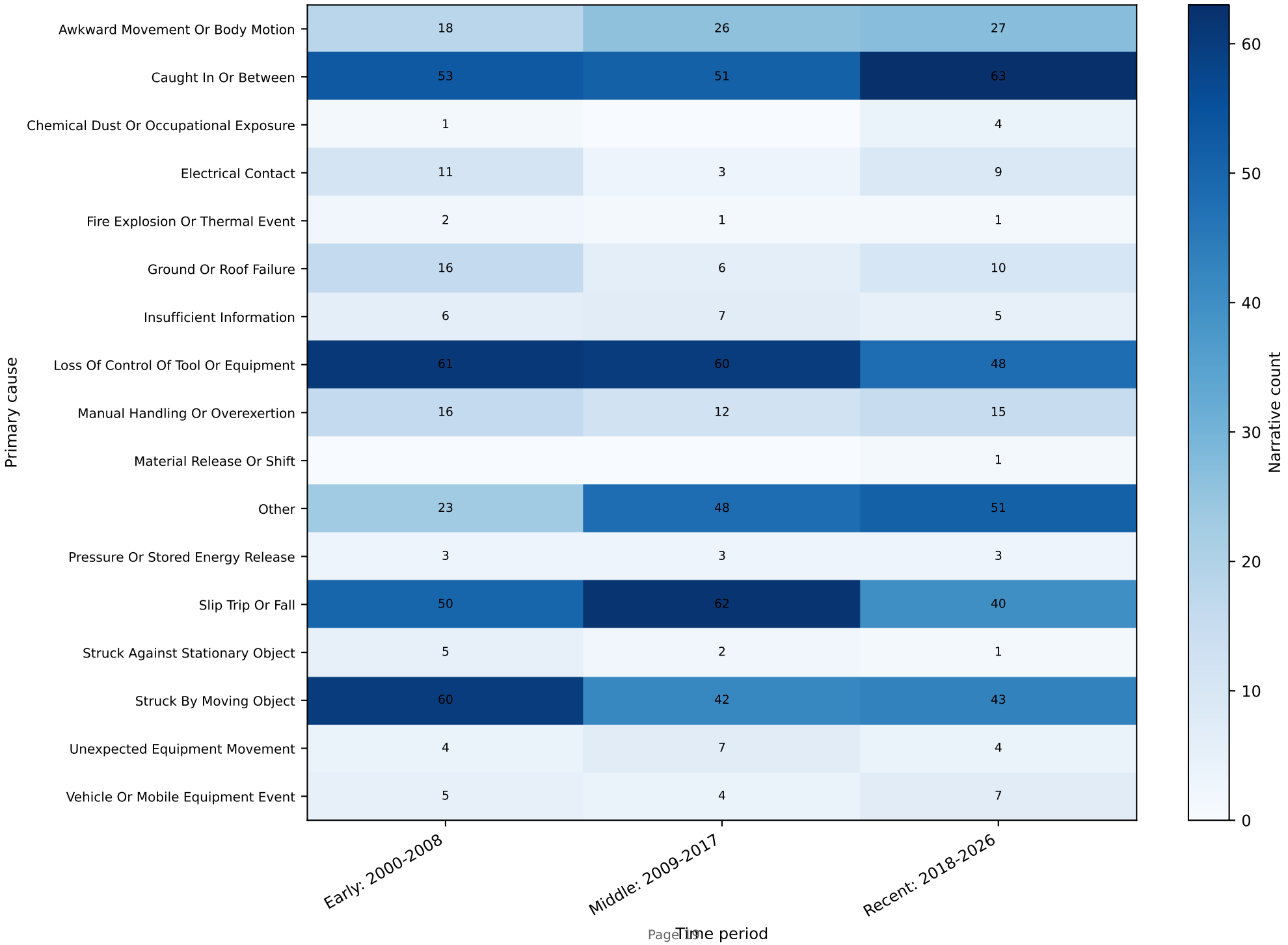
# Prevention Ontology Usage

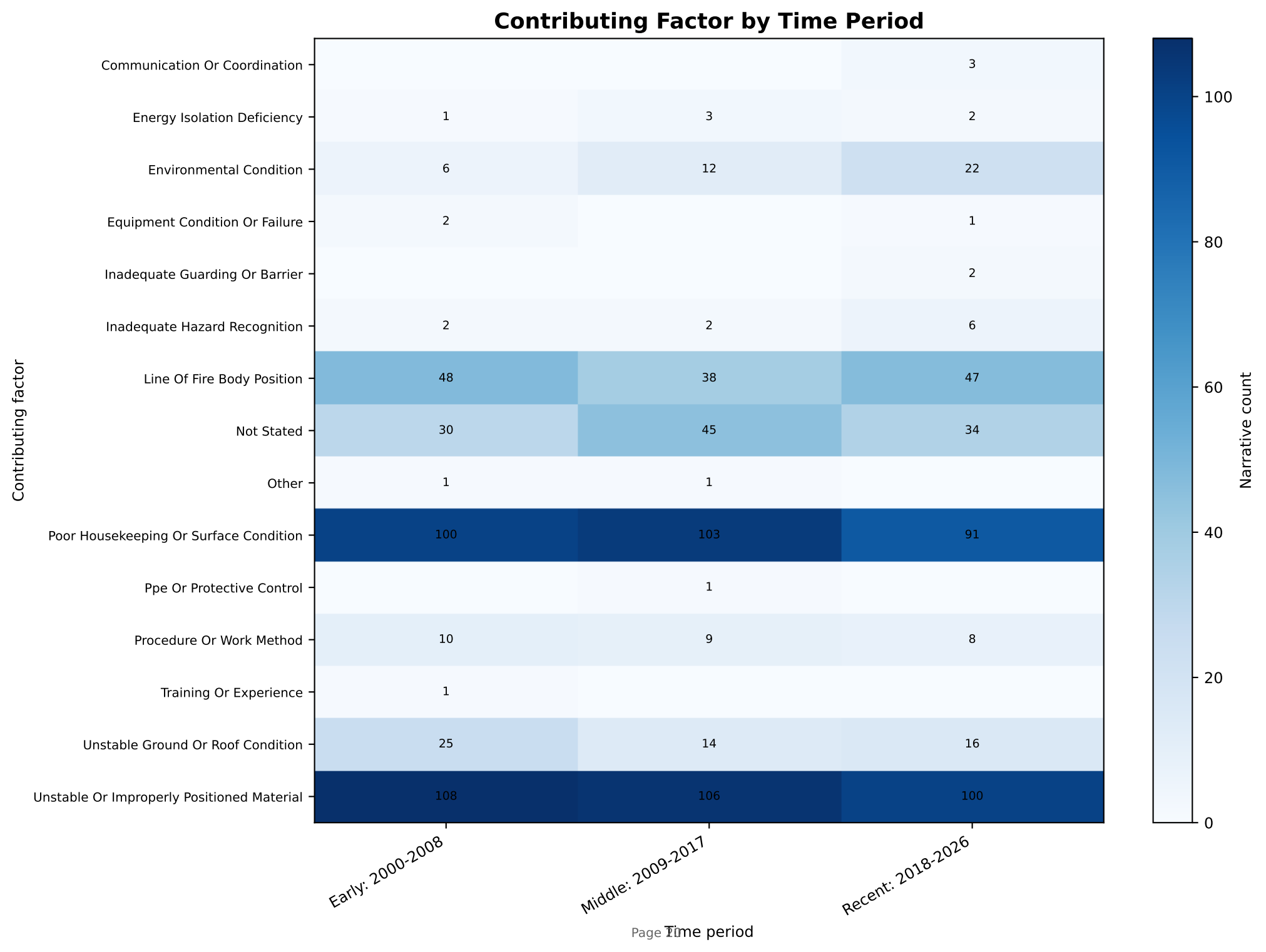
| Category | Count | Percent | Usage Assessment |
|---|---|---|---|
| SECURE_OR_REPOSITION_MATERIAL | 632 | 63.2000 | Potentially overused |
| ESTABLISH_LINE_OF_FIRE_EXCLUSION | 1 | 0.1000 | Very rare |
| ISOLATE_OR_CONTROL_STORED_ENE | 39 | 3.9000 | Used |
| REPAIR_REPLACE_OR_INSPECT_EQUIP | 31 | 3.1000 | Used |
| INSTALL_OR_IMPROVE_GUARDING | 6 | 0.6000 | Used |
| IMPROVE_WORK_SURFACE_OR_HOUS | 127 | 12.7000 | Used |
| IMPROVE_GROUND_CONTROL_OR_SC | 24 | 2.4000 | Used |
| REVISE_PROCEDURE_OR_JOB_PLAN | 13 | 1.3000 | Used |
| CONDUCT_PRE_JOB_HAZARD_REVIEW | 30 | 3.0000 | Used |
| PROVIDE_TARGETED_TRAINING | 33 | 3.3000 | Used |
| IMPROVE_COMMUNICATION_OR_SPOT | 6 | 0.6000 | Used |
| USE_MECHANICAL_HANDLING_ASSIST | 0 | 0.0000 | Unused |
| IMPROVE_PPE_OR_PROTECTIVE_EQUI | 29 | 2.9000 | Used |
| REDESIGN_WORKSPACE_OR_ACCESS | 8 | 0.8000 | Used |
| ENGINEERING_REVIEW_REQUIRED | 4 | 0.4000 | Very rare |
| INVESTIGATION_REQUIRED | 15 | 1.5000 | Used |
| INSUFFICIENT_INFORMATION | 2 | 0.2000 | Very rare |

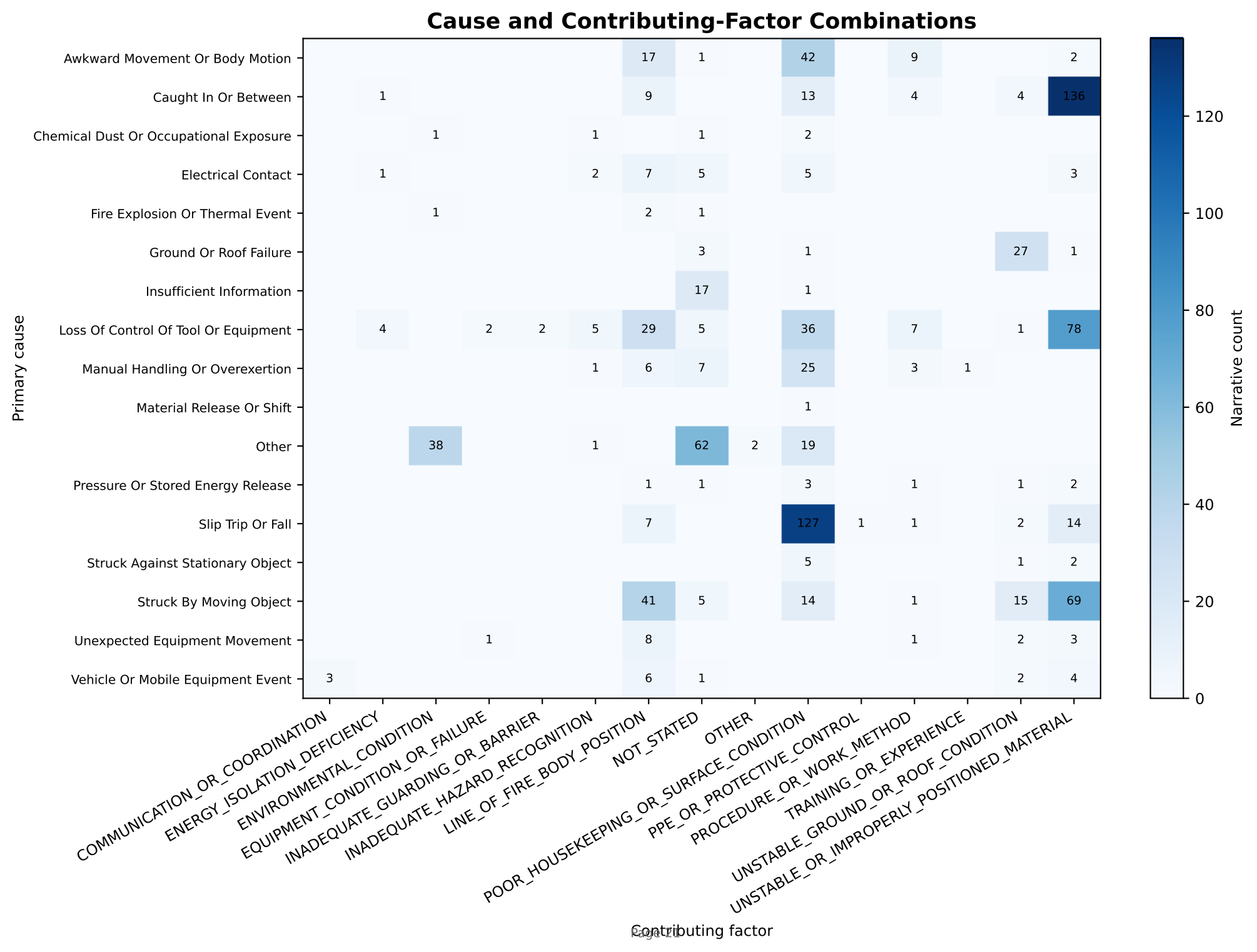

Primary Cause by Injury Severity
Awkward Movement Or Body Motion
Caught In Or Between
Chemical Dust Or Occupational Exposure
Electrical Contact
Fire Explosion Or Thermal Event
Ground Or Roof Failure
Insufficient Information
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Material Release Or Shift
Other
Pressure Or Stored Energy Release
Slip Trip Or Fall
Struck Against Stationary Object
Struck By Moving Object
Unexpected Equipment Movement
Vehicle Or Mobile Equipment Event
Primary cause
21 2 14 4 30
16 31 8 92 20
2 3
1 13 6 1 2
4
3 25 1 2 1
8 8 1 1
46 25 16 44 38
9 10 7 17
1
15 4 72 15 16
1 2 2 2 2
43 17 31 8 53
2 2 2 2
40 48 18 21 18
1 12 1 1
7 7 2
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury severity group
Narrative count
0
20
40
60
80

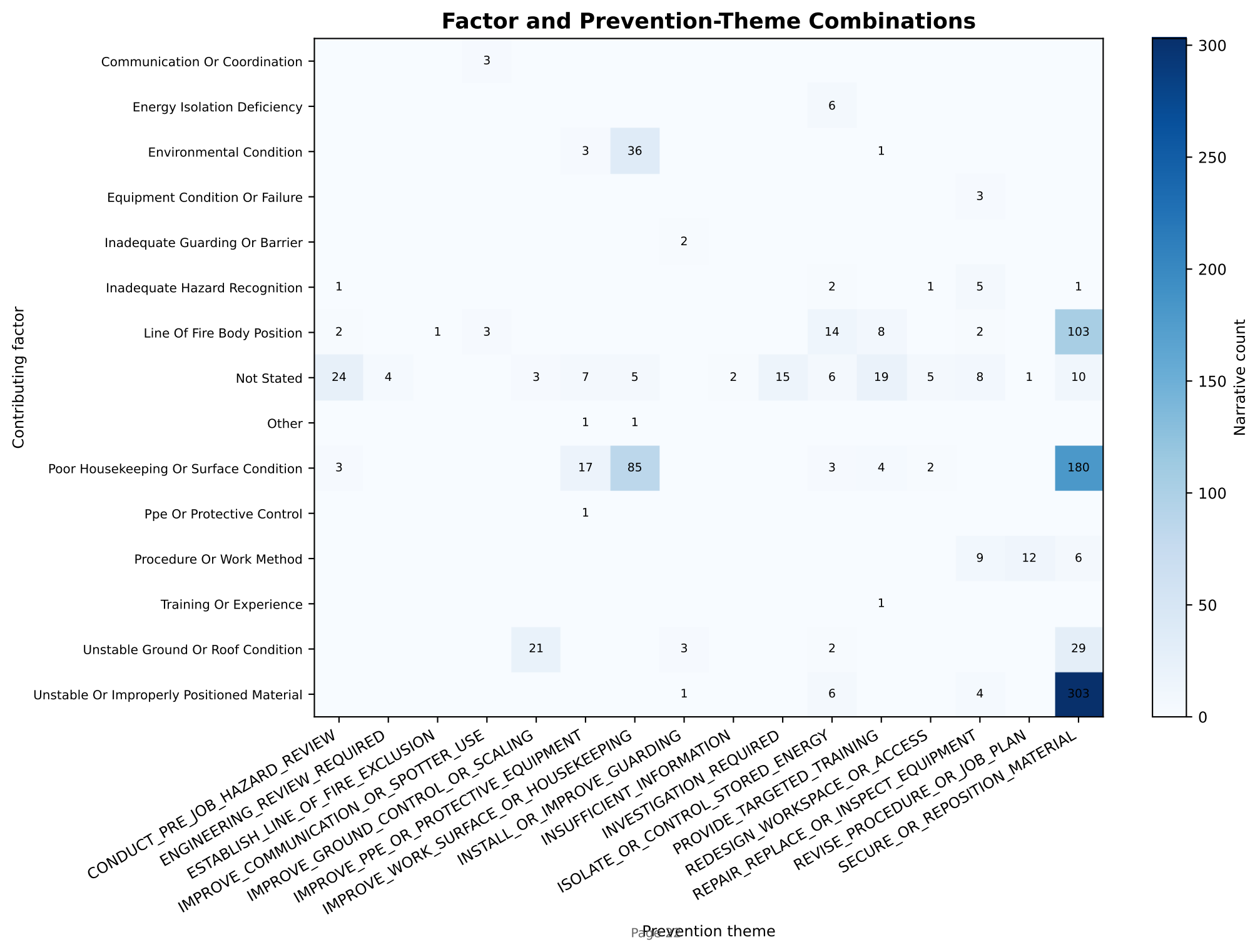

Contributing Factor by Injury Severity
Communication Or Coordination
Energy Isolation Deficiency
Environmental Condition
Equipment Condition Or Failure
Inadequate Guarding Or Barrier
Inadequate Hazard Recognition
Line Of Fire Body Position
Not Stated
Other
Poor Housekeeping Or Surface Condition
Ppe Or Protective Control
Procedure Or Work Method
Training Or Experience
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
Contributing factor
2 1
1 1 4
2 1 37
2 1
1 1
2 2 3 1 2
26 36 21 23 27
17 23 37 18 14
2
78 12 69 28 107
1
10 7 2 2 6
1
7 39 4 4 1
56 75 24 118 41
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury severity group
Narrative count
0
20
40
60
80
100


Prevention Theme by Injury Severity
Conduct Pre Job Hazard Review
Engineering Review Required
Establish Line Of Fire Exclusion
Improve Communication Or Spotter Use
Improve Ground Control Or Scaling
Improve Ppe Or Protective Equipment
Improve Work Surface Or Housekeeping
Install Or Improve Guarding
Insufficient Information
Investigation Required
Isolate Or Control Stored Energy
Provide Targeted Training
Redesign Workspace Or Access
Repair Replace Or Inspect Equipment
Revise Procedure Or Job Plan
Secure Or Reposition Material
Prevention theme
2 2 12 9 5
4
1
3 2 1
1 21 1 1
12 1 12 1 3
20 6 69 3 29
1 3 2
1 1
3 10 1 1
3 18 6 8 4
8 7 6 12
2 2 1 3
7 7 5 4 8
8 2 1 2
136 130 74 160 132
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury-severity group
Narrative count
160
140
120
100
80
60
40
20
0


Primary Cause by Time Period
Awkward Movement Or Body Motion
Caught In Or Between
Chemical Dust Or Occupational Exposure
Electrical Contact
Fire Explosion Or Thermal Event
Ground Or Roof Failure
Insufficient Information
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Material Release Or Shift
Other
Pressure Or Stored Energy Release
Slip Trip Or Fall
Struck Against Stationary Object
Struck By Moving Object
Unexpected Equipment Movement
Vehicle Or Mobile Equipment Event
Primary cause
18 26 27
53 51 63
1 4
11 3 9
2 1 1
16 6 10
6 7 5
61 60 48
16 12 15
1
23 48 51
3 3 3
50 62 40
5 2 1
60 42 43
4 7 4
5 4 7
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
Time period
Narrative count
0
10
20
30
40
50
60


Contributing Factor by Time Period
Communication Or Coordination
Energy Isolation Deficiency
Environmental Condition
Equipment Condition Or Failure
Inadequate Guarding Or Barrier
Inadequate Hazard Recognition
Line Of Fire Body Position
Not Stated
Other
Poor Housekeeping Or Surface Condition
Ppe Or Protective Control
Procedure Or Work Method
Training Or Experience
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
Contributing factor
3
1 3 2
6 12 22
2 1
2
2 2 6
48 38 47
30 45 34
1 1
100 103 91
1
10 9 8
1
25 14 16
108 106 100
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
Time period
Narrative count
100
80
60
40
20
0


Cause and Contributing-Factor Combinations
Primary cause
Awkward Movement Or Body Motion
Caught In Or Between
Chemical Dust Or Occupational Exposure
Electrical Contact
Fire Explosion Or Thermal Event
Ground Or Roof Failure
Insufficient Information
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Material Release Or Shift
Other
Pressure Or Stored Energy Release
Slip Trip Or Fall
Struck Against Stationary Object
Struck By Moving Object
Unexpected Equipment Movement
Vehicle Or Mobile Equipment Event
COMMUNICATION_OR_COORDINATION
ENERGY_ISOLATION_DEFICIENCY
ENVIRONMENTAL_CONDITION
EQUIPMENT_CONDITION_OR_FAILURE
INADEQUATE_GUARDING_OR_BARRIER
INADEQUATE_HAZARD_RECOGNITION
LINE_OF_FIRE_BODY_POSITION
NOT_STATED
OTHER
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
PPE_OR_PROTECTIVE_CONTROL
PROCEDURE_OR_WORK_METHOD
TRAINING_OR_EXPERIENCE
UNSTABLE_GROUND_OR_ROOF_CONDITION
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Contributing factor
Narrative count
0
20
40
60
80
100
120


Factor and Prevention-Theme Combinations
Contributing factor
Communication Or Coordination
Energy Isolation Deficiency
Environmental Condition
Equipment Condition Or Failure
Inadequate Guarding Or Barrier
Inadequate Hazard Recognition
Line Of Fire Body Position
Not Stated
Other
Poor Housekeeping Or Surface Condition
Ppe Or Protective Control
Procedure Or Work Method
Training Or Experience
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
CONDUCT_PRE_JOB_HAZARD_REVIEW
ENGINEERING_REVIEW_REQUIRED
ESTABLISH_LINE_OF_FIRE_EXCLUSION
IMPROVE_COMMUNICATION_OR_SPOTTER_USE
IMPROVE_GROUND_CONTROL_OR_SCALING
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
INSTALL_OR_IMPROVE_GUARDING
INSUFFICIENT_INFORMATION
INVESTIGATION_REQUIRED
ISOLATE_OR_CONTROL_STORED_ENERGY
PROVIDE_TARGETED_TRAINING
REDESIGN_WORKSPACE_OR_ACCESS
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
REVISE_PROCEDURE_OR_JOB_PLAN
SECURE_OR_REPOSITION_MATERIAL
Prevention theme
Narrative count
0
50
100
150
200
250
300

# Automated Review Reasons

| Review Reason | Count |
|---|---|
| Factor-prevention combination is unusual | 531 |
| At least one support level is low | 209 |
| At least one confidence value is low | 154 |
| Cause evidence not found in narrative | 117 |
| Contributing factor is not stated | 100 |
| Factor evidence not found in narrative | 58 |
| Prevention theme requires further review | 12 |
| Narrative may be truncated | 11 |
| Cause is insufficient information | 9 |
| Model response failed validation | 9 |
| Model supplied a reviewer note | 1 |

# Ontology Development Recommendations

1. The schema-failure rate is at or below 5%. Retain the current normalization and retry approach unless individual failures reveal a repeated formatting problem.

2. More than half of the development records require review. Examine the most common review reasons and revise deterministic review rules or ontology definitions before Block 2.

3. More than 15% of records have unusual factor-prevention combinations. Review the compatibility map and determine whether the ontology or action-selection prompt requires refinement.

4. Unused cause categories: ERGONOMIC_OR_REPETITIVE_EXPOSURE. Determine whether these categories are needed, too narrowly defined, or absent from the development sample.

5. Unused factor categories: INADEQUATE_SECUREMENT_OR_RESTRAINT, WORKSPACE_OR_ACCESS_LIMITATION. Determine whether these categories are needed, too narrowly defined, or absent from the development sample.

6. Potentially overused factor categories: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL. Review representative examples for category drift.

7. Unused prevention categories: USE_MECHANICAL_HANDLING_ASSISTANCE. Determine whether these categories are needed, too narrowly defined, or absent from the development sample.

8. Potentially overused prevention categories: SECURE_OR_REPOSITION_MATERIAL. Review representative examples for category drift.

9. Do not freeze the ontology solely from frequency counts. Review representative high-support records, low-support records, alignment flags, and fallback-category records before Block 2.

10. After the Block 1 review is complete, freeze the ontology, system prompt, alias mapping, and compatibility rules. Block 2 should use a separate 2,000-record sample that excludes every ID in the saved development-ID file.

# Representative High-Support Results - 1 of 4

These records completed successfully, were not automatically flagged for review, and had comparatively strong confidence values. They should be inspected to confirm that apparently clean results are substantively correct.

================================================================================
Example 1 | Record 220140060024 | Permanent disability | Middle: 2009-2017
Cause: CAUGHT_IN_OR_BETWEEN | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
On 1/3/14 the employee elected to have his left little finger removed. The finger was injured on 5/28/13 when the employee removed the U-bolts from the axle of a welder trailer and his hand was pinched between the trailer frame and the axle hub. The ligament damage to the finger has not healed properly

Cause evidence: his hand was pinched between the trailer frame and the axle hub
Factor evidence: the U-bolts from the axle of a welder trailer
Prevention rationale: Properly securing the trailer frame and axle would prevent the hand from being caught between them.

================================================================================
Example 2 | Record 220092660006 | Permanent disability | Middle: 2009-2017
Cause: CAUGHT_IN_OR_BETWEEN | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
Knocking timbers, hand was between timber and rip pin, mashed his finger. Partial amputation.

Cause evidence: Knocking timbers, hand was between timber and rip pin, mashed his finger.
Factor evidence: Knocking timbers, hand was between timber and rip pin, mashed his finger.
Prevention rationale: Secure the timber and rip pin to prevent the hand from being caught.

================================================================================
Example 3 | Record 220090260025 | Permanent disability | Middle: 2009-2017
Cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
Closed door on loader and cut end of rt index finger off.

Cause evidence: Closed door on loader and cut end of rt index finger off.
Factor evidence: Closed door on loader and cut end of rt index finger off.

# Representative High-Support Results - 2 of 4

Prevention rationale: Secure the loader doors to prevent accidental closure on fingers.

==========
Example 4 | Record 220091870049 | Permanent disability | Middle: 2009-2017
Cause: CAUGHT_IN_OR_BETWEEN | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
Employee was taking a chunk of shale out of bin when left hand pinky got caught in gate. 1/2" of top part of pinky finger came off.

Cause evidence: Employee was taking a chunk of shale out of bin when left hand pinky got caught in gate.
Factor evidence: 1/2" of top part of pinky finger came off.
Prevention rationale: Secure the gate to prevent fingers from getting caught.

==========
Example 5 | Record 220091130010 | Days away | Middle: 2009-2017
Cause: SLIP_TRIP_OR_FALL | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: HIGH
Review required: False
Review reasons:

Narrative:
Slipped as he stepped off a ladder, twisting his left knee.

Cause evidence: Slipped as he stepped off a ladder, twisting his left knee.
Factor evidence: Slipped as he stepped off a ladder, twisting his left knee.
Prevention rationale: Improving the work surface can prevent slips and falls.

==========
Example 6 | Record 220031710035 | No lost time or other | Early: 2000-2008
Cause: SLIP_TRIP_OR_FALL | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: HIGH
Review required: False
Review reasons:

Narrative:
EE STEPPED ON A ROCK, TWISTED HIS LEFT ANKLE AND BROKE BONE IN HIS LEFT FOOT. THE INJURY REQUIRED SURGERY.

Cause evidence: EE STEPPED ON A ROCK, TWISTED HIS LEFT ANKLE AND BROKE BONE IN HIS LEFT FOOT.
Factor evidence: EE STEPPED ON A ROCK, TWISTED HIS LEFT ANKLE AND BROKE BONE IN HIS LEFT FOOT.
Prevention rationale: Improving the work surface by removing or properly securing rocks can prevent tripping hazards.

# Representative High-Support Results - 3 of 4

```
================================================================================
Example 7 | Record 220181090005 | Restricted or transferred | Recent: 2018-2026
Cause: SLIP_TRIP_OR_FALL | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: HIGH
Review required: False
Review reasons:

Narrative:
Snow and ice accumulation on roadway surface, created a very slippery walkway. The injured employee
slipped and sprained EE's left ankle.

Cause evidence: Snow and ice accumulation on roadway surface, created a very slippery walkway.
Factor evidence: The injured employee slipped and sprained EE's left ankle.
Prevention rationale: Improve the surface condition to reduce slip hazards.

================================================================================
Example 8 | Record 220143240016 | Permanent disability | Middle: 2009-2017
Cause: STRUCK_BY_MOVING_OBJECT | Support: HIGH
Factor: LINE_OF_FIRE_BODY_POSITION | Support: HIGH
Prevention: ESTABLISH_LINE_OF_FIRE_EXCLUSION | Support: HIGH
Review required: False
Review reasons:

Narrative:
Associate stepped in front of a moving train and was struck from behind.

Cause evidence: Associate stepped in front of a moving train and was struck from behind.
Factor evidence: Associate stepped in front of a moving train and was struck from behind.
Prevention rationale: Keep people outside the path of moving trains.

================================================================================
Example 9 | Record 220091700026 | Fatality | Middle: 2009-2017
Cause: STRUCK_BY_MOVING_OBJECT | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
Employee was looking for hydraulic leak and was struck in head by eject all plate. Employee received a
massive head trauma and later became a fatality.

Cause evidence: Employee was looking for hydraulic leak and was struck in head by eject all plate.
Factor evidence: Employee was looking for hydraulic leak and was struck in head by eject all plate.
Prevention rationale: Secure the eject all plate to prevent it from striking employees.

================================================================================
Example 10 | Record 220031750048 | Permanent disability | Early: 2000-2008
Cause: CAUGHT_IN_OR_BETWEEN | Support: HIGH
```

# Representative High-Support Results - 4 of 4

Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
OPERATOR WAS INSTALLING ROOF BOLTS AND MASHED FINGER IN BETWEEN TOP AND ROOF BOLT PLATE.

Cause evidence: OPERATOR WAS INSTALLING ROOF BOLTS AND MASHED FINGER IN BETWEEN TOP AND ROOF BOLT PLATE.
Factor evidence: OPERATOR WAS INSTALLING ROOF BOLTS AND MASHED FINGER IN BETWEEN TOP AND ROOF BOLT PLATE.
Prevention rationale: Secure the roof bolt plate to prevent fingers from being caught.

================================================================================
Example 11 | Record 220151240029 | Days away | Middle: 2009-2017
Cause: CAUGHT_IN_OR_BETWEEN | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
He was putting a piece of angle iron onto another and caught his finger in between, smashing his left ring finger.

Cause evidence: He was putting a piece of angle iron onto another and caught his finger in between, smashing his left ring finger.
Factor evidence: He was putting a piece of angle iron onto another and caught his finger in between, smashing his left ring finger.
Prevention rationale: Secure the angle iron to prevent it from shifting and causing a catch point.

================================================================================
Example 12 | Record 220150830013 | Days away | Middle: 2009-2017
Cause: STRUCK_BY_MOVING_OBJECT | Support: HIGH
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: False
Review reasons:

Narrative:
While sitting on the roof bolter, eating his lunch, a piece of slate fell out of the top and struck him on the back of the neck.

Cause evidence: a piece of slate fell out of the top and struck him on the back of the neck
Factor evidence: a piece of slate fell out of the top
Prevention rationale: Secure the material to prevent pieces from falling.

# Records Requiring Review - 1 of 6

These records were automatically flagged because of missing evidence, low support, fallback categories, truncation, schema problems, or unusual factor-prevention relationships.

================================================================================
Example 1 | Record 220182010006 | No lost time or other | Recent: 2018-2026
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Employee was driving EE's truck with a supervisor. The employee said "something is weird with my eye." The supervisor saw the injured employees left eye start to bleed. Emergency room indicated this was not due to work and cannot be connected to the job.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 2 | Record 220151190057 | Permanent disability | Middle: 2009-2017
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Employee entered entry 1 to follow up on cleaning. 12 - 15' inby the xcut he heard the roof pop, looking up but not seeing anything he went on. A rock 73"x48"x4" fell from between the center bolts, knocked him to the floor and landed on his left foot, fracturing 4 toes and causing the amputation of his pinky toe.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 3 | Record 220001960001 | Fatality | Early: 2000-2008
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
A FATAL ACCIDENT OCCURRED ON OR NEAR THIS CLAIM WHEN A BULLDOZER OVERTURNED ON A STEEP GRADE. ACCORDING TO WHITNESSES THE INDPENDENT CONTRACTOR BECAME OVERLY AGGRESSIVE. ATTEMPTING TO PUSH A B OULDER DOWN A HILL LOSING CONTROL OF THE DOZER. INVETIGATES INDICATE THE DRIVER WAS NOT WEARING A SEAT BELT AT THE TIME OF THE ACCIDENT.

# Records Requiring Review - 2 of 6

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 4 | Record 220171990019 | No lost time or other | Middle: 2009-2017
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Employee fainted due to dehydration.  Immediately regained consciousness.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 5 | Record 220131680015 | Fatality | Middle: 2009-2017
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Employee walked up to the 7-2C header to inspect wipers. The victim was on an elevated catwalk at discharge of the 7-2B conveyor belt to examine the belt wiper. He fell off the end of catwalk on 7-2B belt. Employee was caught between the skirt and the top belt resulting in a fatal injury.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 6 | Record 220191150005 | No lost time or other | Recent: 2018-2026
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Accident occurred in an area of property not mined - individual was camping on property - person was not a miner for Acme to best of Acme knowledge - property leaser gave permission to camp - not verified yet.

Cause evidence:

# Records Requiring Review - 3 of 6

Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 7 | Record 220090280038 | No lost time or other | Early: 2000-2008
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
Welding rebar for wash bay, became overheated. Due to diabetes blood sugar went high, was not able to bring it down.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 8 | Record 220152160002 | Restricted or transferred | Middle: 2009-2017
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
EE was stacking rock in bins off the front of off the chopper and strained his back he came to me and said his back hurt. We made a doctor's appointment for him and he was put on light duty for 10 days.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 9 | Record 220152090004 | No lost time or other | Middle: 2009-2017
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: INVESTIGATION_REQUIRED | Support: LOW
Review required: True
Review reasons: Model response failed validation

Narrative:
The miner suffered from extreme dehydration and muscular cramps. Heat and lack of hydration days before working that day are some of the contributing factors.  The miner received care at a local hospital for dehydration (IV) and considerably helped with muscle cramps as well.  The miner missed 2 full days per doctor's orders.

Cause evidence:

# Records Requiring Review - 4 of 6

Factor evidence:
Prevention rationale: The model response could not be validated.

================================================================================
Example 10 | Record 220021710013 | Days away | Early: 2000-2008
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: PROVIDE_TARGETED_TRAINING | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Cause evidence:
Factor evidence:
Prevention rationale: Targeted training on proper lifting techniques and the risks associated with manual handling could have prevented this incident.

================================================================================
Example 11 | Record 220122750008 | Days away | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Cause evidence:
Factor evidence:
Prevention rationale: Proper protective eyewear could have prevented the foreign object from entering the miner's eye.

================================================================================
Example 12 | Record 220232500006 | No lost time or other | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: ENVIRONMENTAL_CONDITION | Support: MEDIUM
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: MEDIUM
Review required: True
Review reasons: Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

# Records Requiring Review - 5 of 6

Narrative:
EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting. EE was taken to the ER to be seen. EE was treated for heat illness. EE was given a full release at the ER.

Cause evidence:
Factor evidence: EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting.
Prevention rationale: Improving the work environment can help prevent heat-related illnesses.

================================================================================
Example 13 | Record 220250210029 | No lost time or other | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE came to the shop and said side was hurting.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could help identify and address potential issues before they cause pain.

================================================================================
Example 14 | Record 220162160052 | No lost time or other | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE was working outside using a 20lb sledge hammer to loosen material that was stuck inside the chute. EE went in to cool off. About 10 minutes after EE tried calling to get ahold of someone because EE stated was cramping up and skin was hot clammy and EE was shivering.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could have identified the need for rest breaks and monitoring for signs of heat-related illness.

================================================================================
Example 15 | Record 220133640050 | Fatality | Middle: 2009-2017

# Records Requiring Review - 6 of 6

Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: MEDIUM
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: MEDIUM
Review required: True
Review reasons: Cause is insufficient information; Cause evidence not found in narrative; At least one confidence value is low; At least one support level is low

Narrative:
Victim was last observed accessing the stern if barge NBN-611. Shortly after, he was noticed to be missing. Snow was present on the barge. Damage or impairment is unknown at this time.

Cause evidence:
Factor evidence: Snow was present on the barge.
Prevention rationale: Improving the work surface by removing snow could prevent similar incidents.

# Factor-Prevention Alignment Flags - 1 of 5

These records contain factor-prevention combinations outside the current compatibility map. Review whether the model chose an inappropriate prevention theme or whether the compatibility map is too restrictive.

==========
Example 1 | Record 220253350005 | Restricted or transferred | Recent: 2018-2026
Cause: STRUCK_BY_MOVING_OBJECT | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual

Narrative:
While standing at the control panel of a portable screen a large rock fell from the top surface and struck the top of EE's hard hat.  The hard hat shell contacted EE's head, causing a cut to EE's scalp. The impact knocked EE down and caused pain to EE's back and right shoulder.

Cause evidence: a large rock fell from the top surface and struck the top of EE's hard hat.
Factor evidence: The hard hat shell contacted EE's head, causing a cut to EE's scalp.
Prevention rationale: Secure the top surface to prevent rocks from falling and injuring workers.

==========
Example 2 | Record 220001640065 | Restricted or transferred | Early: 2000-2008
Cause: STRUCK_BY_MOVING_OBJECT | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual

Narrative:
EMPLOYEE WAS APPLYING STOPPING SEALANT WITH A SPRAY MACHINE WHEN APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.

Cause evidence: APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.
Factor evidence: APPLICATION HOSE SLIPPED THROUGH HIS HANDS
Prevention rationale: Secure the application hose to prevent it from slipping and causing sealant to strike the employee.

==========
Example 3 | Record 220021710013 | Days away | Early: 2000-2008
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: PROVIDE_TARGETED_TRAINING | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:

# Factor-Prevention Alignment Flags - 2 of 5

EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Cause evidence:
Factor evidence:
Prevention rationale: Targeted training on proper lifting techniques and the risks associated with manual handling could have prevented this incident.

================================================================================
Example 4 | Record 220090090045 | Restricted or transferred | Early: 2000-2008
Cause: MANUAL_HANDLING_OR_OVEREXERTION | Support: HIGH
Factor: LINE_OF_FIRE_BODY_POSITION | Support: LOW
Prevention: PROVIDE_TARGETED_TRAINING | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual; At least one support level is low

Narrative:
Employee was carrying a battery from the back of the truck to the track mobile. As he walked accross the railroad tracks, he felt pain on left side of his back.

Cause evidence: Employee was carrying a battery from the back of the truck to the track mobile.
Factor evidence: As he walked across the railroad tracks, he felt pain on left side of his back.
Prevention rationale: Proper training can help employees recognize and avoid overexertion during manual handling tasks.

================================================================================
Example 5 | Record 220122750008 | Days away | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Cause evidence:
Factor evidence:
Prevention rationale: Proper protective eyewear could have prevented the foreign object from entering the miner's eye.

================================================================================
Example 6 | Record 220032580086 | No lost time or other | Early: 2000-2008
Cause: AWKWARD_MOVEMENT_OR_BODY_MOTION | Support: HIGH
Factor: LINE_OF_FIRE_BODY_POSITION | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True

# Factor-Prevention Alignment Flags - 3 of 5

Review reasons: Factor-prevention combination is unusual

Narrative:
EMPLOYEE WAS PACKING TYPE S ON PORTLAND WHEN TWO BAGS FELL OFF PACKING MACHINE ONTO THE CONVEYOR BELT BELOW. EMPLOYEE TRIED TO HOLD BACK ONE OF THE BAGS WITH HIS RIGHT FOOT TO PREVENT THE TWO BAGS FROM PLUGGING THE TRANSFER POINT. WHILE STRETCHING DOWN AND EXTENDING LEG TO HOLD BACK THE BAGS, HE FELT A POP IN LOWER BACK.

Cause evidence: EMPLOYEE TRIED TO HOLD BACK ONE OF THE BAGS WITH HIS RIGHT FOOT TO PREVENT THE TWO BAGS FROM PLUGGING THE TRANSFER POINT. WHILE STRETCHING DOWN AND EXTENDING LEG TO HOLD BACK THE BAGS, HE FELT A POP IN LOWER BACK.
Factor evidence: EMPLOYEE TRIED TO HOLD BACK ONE OF THE BAGS WITH HIS RIGHT FOOT TO PREVENT THE TWO BAGS FROM PLUGGING THE TRANSFER POINT. WHILE STRETCHING DOWN AND EXTENDING LEG TO HOLD BACK THE BAGS, HE FELT A POP IN LOWER BACK.
Prevention rationale: Securing the bags to prevent them from falling would reduce the need for the employee to stretch and extend their leg.

==========
Example 7 | Record 220023600016 | Restricted or transferred | Early: 2000-2008
Cause: SLIP_TRIP_OR_FALL | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual

Narrative:
WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT. AFTER 2 EXAMS A MRI WAS DONE AND DETERMINED THAT MINER HAD A HAIRLINE FRACTURE ON 12-19-02. THIS BECAME REPORTABLE ON DECEMBE R 19, 2002.

Cause evidence: WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT.
Factor evidence: WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT.
Prevention rationale: Improving housekeeping and securing materials can prevent slips and falls.

==========
Example 8 | Record 220232500006 | No lost time or other | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: ENVIRONMENTAL_CONDITION | Support: MEDIUM
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: MEDIUM
Review required: True
Review reasons: Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting. EE was taken to the ER to be seen. EE was treated for heat illness. EE was given a full release at the ER.

Cause evidence:
Factor evidence: EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting.

# Factor-Prevention Alignment Flags - 4 of 5

Prevention rationale: Improving the work environment can help prevent heat-related illnesses.

================================================================================
Example 9 | Record 220250210029 | No lost time or other | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative;
Factor-prevention combination is unusual; At least one confidence value is low; At least one support
level is low

Narrative:
EE came to the shop and said side was hurting.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could help identify and address potential
issues before they cause pain.

================================================================================
Example 10 | Record 220162160052 | No lost time or other | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative;
Factor-prevention combination is unusual; At least one confidence value is low; At least one support
level is low

Narrative:
EE was working outside using a 20lb sledge hammer to loosen material that was stuck inside the chute. EE
went in to cool off. About 10 minutes after EE tried calling to get ahold of someone because EE stated
was cramping up and skin was hot clammy and EE was shivering.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could have identified the need for rest breaks
and monitoring for signs of heat-related illness.

================================================================================
Example 11 | Record 220192830020 | Restricted or transferred | Recent: 2018-2026
Cause: AWKWARD_MOVEMENT_OR_BODY_MOTION | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: HIGH
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual

Narrative:
EE was carrying a piece of roof screen beside the roof bolter. EE lost footing and twisted knee.

# Factor-Prevention Alignment Flags - 5 of 5

Cause evidence: EE lost footing and twisted knee.
Factor evidence: EE was carrying a piece of roof screen beside the roof bolter.
Prevention rationale: Securing the roof screen would prevent EE from losing footing.

================================================================================
Example 12 | Record 220262470030 | Permanent disability | Recent: 2018-2026
Cause: MANUAL_HANDLING_OR_OVEREXERTION | Support: HIGH
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | Support: LOW
Prevention: SECURE_OR_REPOSITION_MATERIAL | Support: HIGH
Review required: True
Review reasons: Factor-prevention combination is unusual; At least one support level is low

Narrative:
Lifting, pulling, pushing equipment (hoses and pumps) to and from a blasting area to prevent damage. Which caused a hernia for the miner.

Cause evidence: Lifting, pulling, pushing equipment (hoses and pumps) to and from a blasting area to prevent damage.
Factor evidence: Which caused a hernia for the miner.
Prevention rationale: Improving the handling methods and securing the equipment can reduce the risk of overexertion.

# NOT_STATED Contributing-Factor Examples - 1 of 5

These records were assigned NOT_STATED as the contributing factor. Review whether the narratives genuinely omit factor information or whether the prompt is abstaining too often.

================================================================================
Example 1 | Record 220021710013 | Days away | Early: 2000-2008
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: PROVIDE_TARGETED_TRAINING | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Cause evidence:
Factor evidence:
Prevention rationale: Targeted training on proper lifting techniques and the risks associated with manual handling could have prevented this incident.

================================================================================
Example 2 | Record 220122750008 | Days away | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Cause evidence:
Factor evidence:
Prevention rationale: Proper protective eyewear could have prevented the foreign object from entering the miner's eye.

================================================================================
Example 3 | Record 220250210029 | No lost time or other | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

# NOT_STATED Contributing-Factor Examples - 2 of 5

Narrative:
EE came to the shop and said side was hurting.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could help identify and address potential issues before they cause pain.

================================================================================
Example 4 | Record 220162160052 | No lost time or other | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE was working outside using a 20lb sledge hammer to loosen material that was stuck inside the chute. EE went in to cool off. About 10 minutes after EE tried calling to get ahold of someone because EE stated was cramping up and skin was hot clammy and EE was shivering.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could have identified the need for rest breaks and monitoring for signs of heat-related illness.

================================================================================
Example 5 | Record 220231930009 | Permanent disability | Recent: 2018-2026
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
Employee was bending over moving tools when they felt a sharp pain in their abdominal area. After further evaluation at ER, determined that employee had a unilateral inguinal hernia.

Cause evidence:
Factor evidence:
Prevention rationale: Conducting a pre-job hazard review could have identified the risk of abdominal strain during bending over moving tools.

================================================================================
Example 6 | Record 220092510049 | Permanent disability | Middle: 2009-2017

# NOT_STATED Contributing-Factor Examples - 3 of 5

Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative;
Factor-prevention combination is unusual; At least one confidence value is low; At least one support
level is low

Narrative:
While filling the rock dusting maachine with rock dust, Injured complained with pain to his groin area.

Cause evidence:
Factor evidence:
Prevention rationale: A pre-job hazard review could have identified potential issues and prevented the
injury.

================================================================================
Example 7 | Record 220113120011 | Permanent disability | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative;
Factor-prevention combination is unusual; At least one confidence value is low; At least one support
level is low

Narrative:
Employee was removing a rock from the #2 take-up when he felt a pain in his shoulder, back and abdomen.
He continued to work until 10-3-11 when he started missing work. Workers Comp was looki ng to see if a
pre-existing condition existed and the Safety Dept. was notified on 11-2-11 that the accident did not
involve a pre-existing condition. Diagnosed with a hernia.

Cause evidence:
Factor evidence:
Prevention rationale: A pre-job hazard review could have identified the potential for injury from heavy
lifting or repetitive motion.

================================================================================
Example 8 | Record 220112590016 | No lost time or other | Middle: 2009-2017
Cause: OTHER | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | Support: LOW
Review required: True
Review reasons: Contributing factor is not stated; Cause evidence not found in narrative;
Factor-prevention combination is unusual; At least one confidence value is low; At least one support
level is low

Narrative:
Employee was working on revolving frame and became ill.  He was taken for medical treatment and treated
for heat exhaustion.

# NOT_STATED Contributing-Factor Examples - 4 of 5

Cause evidence:
Factor evidence:
Prevention rationale: Improving work surface conditions could help prevent heat-related illnesses.

================================================================================
Example 9 | Record 220032100028 | Restricted or transferred | Early: 2000-2008
Cause: ELECTRICAL_CONTACT | Support: HIGH
Factor: NOT_STATED | Support: LOW
Prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
EE WAS ROCK BOLTING AND A SMALL PIECE OF METAL GOT IN HIS EYE.

Cause evidence: EE WAS ROCK BOLTING AND A SMALL PIECE OF METAL GOT IN HIS EYE.
Factor evidence:
Prevention rationale: Proper protective eyewear could have prevented the eye injury.

================================================================================
Example 10 | Record 220103070019 | No lost time or other | Middle: 2009-2017
Cause: MANUAL_HANDLING_OR_OVEREXERTION | Support: HIGH
Factor: NOT_STATED | Support: LOW
Prevention: REPAIR_REPLACE_OR_INSPECT_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
Back strain while trying to lift a blower onto a frame.

Cause evidence: Back strain while trying to lift a blower onto a frame.
Factor evidence:
Prevention rationale: Improving the lifting method could reduce the risk of back strain.

================================================================================
Example 11 | Record 220082950042 | Fatality | Early: 2000-2008
Cause: INSUFFICIENT_INFORMATION | Support: LOW
Factor: NOT_STATED | Support: LOW
Prevention: ENGINEERING_REVIEW_REQUIRED | Support: HIGH
Review required: True
Review reasons: Cause is insufficient information; Contributing factor is not stated; Prevention theme requires further review; At least one confidence value is low; At least one support level is low

Narrative:
The Employee was on the surface of a clean dry barge in daylight hours with no movement and light wind. The Ee passed through the #2 port hatch opening and made contact with the empty hopper floor. Cause is unknown. Ee was installing a dust prevention tarp across the port hatch opening before the incident. The

# NOT_STATED Contributing-Factor Examples - 5 of 5

operator contests MSHA jurisdiction.

Cause evidence: Cause is unknown.
Factor evidence:
Prevention rationale: An engineering review is needed to determine the cause of the incident.

================================================================================
Example 12 | Record 220112290034 | Fatality | Middle: 2009-2017
Cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT | Support: HIGH
Factor: NOT_STATED | Support: LOW
Prevention: REPAIR_REPLACE_OR_INSPECT_EQUIPMENT | Support: HIGH
Review required: True
Review reasons: Contributing factor is not stated; Factor-prevention combination is unusual; At least one confidence value is low; At least one support level is low

Narrative:
The victim was leaning over a base lift jack housing. The molded casing catastrophically failed striking the victim in the chest from unknown/undetectable reasons. Cause was catastrophic failure of the housing. Compliance with rules/regs, protective items not a factor, miner proficiency may be a factor, mining equip/systems may be a factor.

Cause evidence: The molded casing catastrophically failed striking the victim in the chest from unknown/undetectable reasons. Cause was catastrophic failure of the housing.
Factor evidence:
Prevention rationale: Inspect and repair the base lift jack housing to prevent catastrophic failures.

# Block 1 Conclusions and Decision Criteria

Block 1 is complete when the development report has been reviewed and the project has made final decisions regarding:

1. Cause categories and definitions.

2. Contributing-factor categories and definitions.

3. Prevention-theme categories and definitions.

4. Category aliases and schema-repair rules.

5. Evidence requirements.

6. Factor-prevention compatibility rules.

7. Automated human-review conditions.

8. Treatment of occupational illnesses, noninjury reports, and natural or nonoccupational events.

After those decisions are made, the ontology and prompt should be frozen. Block 2 should then select 2,000 unseen narratives while excluding all IDs stored in the Block 1 development-ID file.

Block 1 frequency distributions should not be interpreted as population prevalence because the development sample is balanced by severity.

Final accuracy will be measured later using the independently reviewed 100-record sample from Block 2.

# Technical Appendix

```
Model: Qwen/Qwen2.5-7B-Instruct
Prompt version: MSHA_BLOCK1_ONTOLOGY_DEVELOPMENT_V1.0
GPU: Tesla T4
Development records: 1,000
Cause categories: 18
Factor categories: 17
Prevention categories: 17
Minimum narrative words: 8
Truncation threshold: 383
Maximum retries: 3
Maximum new tokens: 550
Random state: 42

PDF report: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block1_Ontology_Development_Report.pdf
Checkpoint: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block1_Development_Checkpoint.jsonl
Development IDs: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block1_Development_Record_IDs.json
```